Reasoning Depth versus Orchestration in Mathematical AI Harnesses

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Abstract

Mathematical AI systems increasingly wrap language models in harnesses that allocate work among model calls, critics, tools, and persistent state. We ask whether a fixed inference budget is better spent on longer coherent reasoning or external orchestration. We compare direct long reasoning, a multi-agent harness, and Elenx, a reasoning-heavy system that preserves durable state and independently verifies candidate proofs. Our evaluation measures verified solutions as a function of total dollars and decomposes spend into reasoning, coordination, verification, and retries.

1 Introduction

Accuracy-only comparisons hide where inference budget goes. Elenx keeps search inside long reasoning episodes, while a small kernel records exact candidates, calls, and verifier-bound verdicts. We ask whether this reasoning depth is a better use of budget than adding orchestration. Our contributions are a matched-cost protocol, an auditable comparison, and a budget decomposition.

2 Systems and hypothesis

We compare a direct long reasoning call, Danus, the author's earlier multi-agent harness, and Elenx. All arms receive the same base model, statement, tools, verifier, and dollar budget. We hypothesize that coherent reasoning can improve the capability–cost frontier on difficult research-style problems, but may be dominated when the base model is already strong.

3 Evaluation

Tasks are split into development and held-out sets. A success is a complete proof or disproof accepted by an independent verifier. We run paired trials at fixed dollar budgets and report verified tasks solved, cost, reasoning tokens, coordination tokens, calls, retries, and elapsed time. Every paid leaf records provider, model, reasoning level, usage, and billed cost.

4 Initial evidence

An Elenx campaign on IMO 2011 Problem 2 contains 26 model calls, 13 tool calls, four candidate proofs, and four independent verdicts. Three candidates were rejected with localized objections to load-bearing lemmas. The fourth received PASS after a self-contained proof covering all cardinalities. A separate Danus run provides the motivating cost contrast: Danus allocated only 8.3% of cost to reasoning (46% of output tokens), compared with 24.5% and 56% for our coordinator. The underlying run exceeded $2,000 while failing to reach a sufficient verified resolution; we will cite its durable accounting record in the supplement. These are audited case studies, not yet a statistically powered comparison.

5 Results and ablations

Table 1: Matched-budget results (to be completed).

System	Budget	Verified	Cost/pass
Direct reasoning	B_1	–	–
Multi-agent harness	B_1	–	–
Elenx	B_1	–	–

We ablate continuation, durable state, independent verification, and long single-episode reasoning. A system is better when it solves at least as many tasks for no greater cost.

6 Discussion

A negative result would identify orchestration overhead as a measurable design cost; a positive result would support long coherent reasoning as a useful harness primitive. Claims will remain scoped to the tested benchmark, model snapshot, and verifier.

References

[1] Anonymous. Elenx kernel specification, 2026.